MATH 202 Your name: _____

**Learning target D4, version 1**

Find general solutions to separable differential equations, and use initial values to determine the value of the constant of integration.

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Suppose that a cylindrical water tank with a hole in the bottom is filled with water. The water, of course, will leak out and the height of the water will decrease. Let $h(t)$ denote the height of the water. A physical principle called *Torricelli's Law* implies that the height decreases at a rate proportional to the square root of the height: *

$$\frac{dh}{dt} = -k\sqrt{h}, \quad k > 0.$$

1. Suppose you have a tank for which the height decreases at a rate of 20 inches per minute when the water is filled to a depth of 100 inches. Find the value of $k$.

2. Now solve the differential equation.

3. Finally, use the initial condition $h(0) = 100$ to find the value of the constant of integration.

*Conservation of energy: the kinetic energy of water leaving the tank is equal to the change in GPE.